

Review of Optimization of Power Dispatch in Renewable Energy System

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Abstract—With increasing complexity and unpredictability in power systems, optimization of power dispatch is now more important than ever. It is imperative to overcome the challenges brought about by the rise of renewable energy sources (RES) and electric vehicles (EVs) and to capitalize on the opportunities that they present. The aim of this paper is to give an overview of the multifarious optimization tools and techniques employed to manage power dispatch. The paper then goes on to discuss the advantages and limitations of the optimization techniques and the applications that are suitable for them. Future trends and necessary next steps in the field of power dispatch have also been deliberated.

Index Terms—optimization, power dispatch, renewable energy

I. INTRODUCTION

In recent years, renewable energy sources (RES) such as photovoltaic systems (PV) – representing solar energy – and wind turbines are gaining more attention because of climate change and air pollution issues. Apart from having lower carbon footprint, RES have another significant advantage compared to traditional generation: their modularity. This means that they can be placed near or at the place where electricity is consumed, reducing transmission losses [1]. However, wide adoption of solar and wind energy still faces major challenges. Firstly, both wind and solar power are non-dispatchable: power generation from these sources cannot be adjusted at will, even if it is absolutely necessary to do so. Secondly, although wind and solar energy are already cost competitive in places with some of the highest solar and wind potential, solar panels and wind turbines remain a costly investment in many parts of the world [2]. Moreover, more than 90% of the energy demand growth until 2035 will come from the developing countries [3], whose purchasing power hinders the uptake of renewable energy.

The first problem can be solved by adding an energy storage device to the renewable energy system to smoothen the output power from RES, although this will aggravate the second problem. The second problem is ameliorated with time as prices continue to decrease with increasing production and deployment of RES. More importantly, controlling and optimizing the power dispatch in the renewable energy system can also mitigate the two problems, by tuning the system to accommodate for the fluctuation of RES and by minimizing the cost of running the systems, respectively.

Power dispatch problem, also known as generation scheduling problem, is a classic problem consisting of a unit commitment [4] and economic dispatch. Unit commitment problem seeks to find the optimum generators' up and down time to fulfill the power demand in the near future. Economic dispatch is about finding the optimum operating point of these generators to minimize the fuel cost without changing the on/off setting.

In the past, generation scheduling were usually conducted a day ahead and assumed perfect prediction of future electricity demand. However, with the introduction and the increasing importance of distributed RES in power system, the power dispatch problem has become even more complex. There is a need to predict and take into account the availability of these renewable resources in the future to optimize the system meaningfully. Without such optimization, demand outage may occur because of the mismatch in electricity demand and supply. Consequently, reliability of the system is reduced and energy cost increased.

Therefore, optimization of power dispatch is more important than ever. This work aims to give an overview of the optimization tools and algorithms employed in generation scheduling of renewable energy systems. In this review, different types of algorithms proposed to tackle varying issues in power dispatch are analyzed. The future trends and subsequent work required in the field are also discussed.

Most of the literature reviewed in this work deals with systems of smaller scale, i.e. distribution systems or microgrids rather than transmission systems, as the adoption of new technologies and paradigm will happen at smaller scales first. As such, thermal generators are not normally present and the unit commitment problem becomes less important. It is then more important to account for the variability in the RES.

Optimization techniques can be broadly classified into exact and heuristic optimization. On the one hand, exact optimization guarantees that an optimum solution will be reached in a finite time through analytical approach. However, when the problem is NP-hard, the algorithm may need an exponential time – which may be infeasible for practical purposes – to find the optimum solution [5]. On the other hand, heuristic optimizations give no guarantee that an optimum solution will be found. Nevertheless, it will give 'good enough' solutions

within a 'reasonable' time through trial and error. Examples from the literature, advantages and disadvantages of each optimization technique will be discussed in later sections.

The paper is organized as follows. Section II gives a brief explanations on the tools that are frequently used in conjunction with the optimization algorithms. Section III and IV then introduce and examine exact and heuristic optimization respectively. Section V shows how the two branches of optimization can be incorporated. In Section VI, the various optimization tools will be discussed from algorithm and system perspective. This review is concluded in Section VII. Table I on page 7 is available as a summary to the literature reviewed in this paper.

II. OPTIMIZATION TOOLS AND PLATFORM

To understand the state of the art optimization techniques, it is important to understand the computational intelligence that is frequently utilized together with the optimization algorithms.

A. Artificial Neural Network (ANN)

ANN is a computational technique originally developed for pattern recognition by Rosenblatt in 1958 [6]. Since then, ANN has evolved and has been especially useful in applications that involve prediction [7]. As the name suggests, ANN mimics how the human brain works. It is composed of layers of neurons, namely the input layer, output layer, and hidden layer(s) in between. The neurons in each layer are interconnected with other neurons in the adjacent layer with different adaptive weights. Similar to how a child learns about his surrounding with guidance from his parents, ANN requires training with large data sets to learn the 'correct' or desired outputs given the inputs. The learning process changes the weights between the links. In designing ANN, the number of hidden layers and the number of neurons in each layer are some of the variables which need to be considered.

In the field of energy management, ANN is particularly useful in forecasting the availability of RES and electricity price. With enough training, it can generate more accurate case studies in shorter time than simulations [8].

B. Fuzzy Logic

Fuzzy logic is a decision making tool that is regularly applied when there are uncertainty and ambiguity in the system data and inputs [9]. The process of fuzzy logic can be approximated into three steps. First, the input values are processed by membership functions and turned into fuzzy values. Based on the fuzzified input, the fuzzy output functions are then computed according to a set of specifically crafted rules. Lastly, the fuzzy output functions are defuzzified into crisp information for decision making. Defining the membership functions and set of rules is crucial for the fuzzy logic to be meaningful; and guidelines on the methodology can be found in [10]. As an example of its application, in [11], the authors took the electricity pricing, available generation of DERs, load demand and the state-of-charge of energy storage as the inputs to fuzzy logic and decided the amount of power to be bought or sold to the grid as the output using predefined membership functions.

C. Multiagent System (MAS)

MAS is a platform where each component in the system can be represented as an agent, each with its own objective. Agents are autonomous, social, reactive and proactive, meaning that agents can control their actions without human inputs, interact with each other and with humans, adapt to changes in the environment, and take initiatives to achieve their objectives, respectively [12]. By employing MAS architecture, data gathering and decision making of the power system can be decentralized and will be more robust against single-point failure. A more detailed review on the MAS architecture, development platforms as well as applications in microgrid can be found in [13] and [14].

III. EXACT OPTIMIZATION

Exact optimization requires the objective's and constraints' functions to be known quantitatively. The classifications of the exact optimization are not simple as there are overlaps within these classifications. Exact optimization can be classified as deterministic, stochastic, and robust, depending on how uncertainties are handled. It can also be classified as convex and non-convex optimizations depending on the nature of the objective function and of the constraints.

A. Deterministic Optimization (DO)

DO considers all the variables in the system to be known perfectly, with no uncertainty in the system [15]. Even though in real power system this is never the case, DO is still very useful in providing a solution to generation scheduling problems quickly, as it doesn't have the added complexity of stochastic and robust optimization. DO can also act as a performance upper bound for other optimization techniques if we are assumed to have perfect information about the system.

In [16], Wang, et al. designed a distributed convex optimization in a microgrid with PV, curtailable loads, energy storage and EVs. The objective was to minimize the power taken from the grid as well as the regulation cost to smoothen the power profile of the microgrid. By using the alternating direction method of multipliers (ADMM), the authors distributed the optimization of each problem to DER. Each DER then calculated an optimum power scheduling that solved the local optimization. These values were sent back to a collector that regulated all the values iteratively to balance the power. Subsequently, the solution from ADMM at each time step was employed by the model predictive control (MPC) to control the DER over a finite time horizon. The problem was iterated for each time step to consolidate changes in the environment and to increase the robustness of the algorithm. Thus, each DER only needed to know its own state and its historical power profiles to determine its control actions. Parameters for PV and EVs were predicted using the historical data and the simulation was run at a 15 minute interval for 3 days. ADMM performed only 1.5% worse than a centralized controller and was more robust to communication failures compared to centralized algorithm.

At the household level, Nottrott, et al. [17] designed a linear programming (LP) to optimize grid-connected PV-battery systems using continuous load and solar forecasting. The objective was to minimize electricity taken from the grid, especially by reducing the peak net load. First, a peak load reduction target was set and optimization was conducted when the battery energy capacity was less than the amount of energy necessary to achieve this target (such that simple battery discharge is insufficient). When compared with other methods that do not use forecasting, the bill savings and consequently the value of the battery were increased using the optimized dispatch. Even so, battery purchase for bill savings is still not yet viable economically in the case of San Diego, when the Li-ion battery costs \$1000/kWh, but is increasingly so at lower prices.

One of the ways to incorporate and tackle uncertainties – without the complexity of stochastic and robust optimization – in DO is by first developing an off-line deterministic optimization problem and then forming the online algorithm by adding arbitrary errors to the uncertain parameters [18].

B. Stochastic Optimization (SO)

SO differs from DO as it recognizes that there is an underlying randomness in the variables and/or objective function of the system [19]. Using historical data, the variables are assumed to follow a known distribution, which can be updated when more data becomes available. SO approximates real systems more accurately than its deterministic counterpart but suffers from dependency on historical data as well as complex modeling.

Kekatos, et al. in [20] utilized reactive power generated from PV to provide voltage support in distribution system. The objective function, to minimize average power loss, was relaxed into SOCP (second-order cone programming, a convex optimization). The active and reactive power were modeled as stochastic processes with a probability density function. Estimates of reactive power that minimizes the problem was iteratively found using the Lagrange multipliers as soon as new injections data became available. 47-bus microgrid and residential IEEE 123-bus feeder were used as test cases, in which the algorithm was shown to perform better than the deterministic approach. The stochastic scheme was able to track solar and load variation at one minute period. This paper has provided a stochastic control framework that can be extended to other types of problems such as minimizing voltage deviation.

Meanwhile in [21], Gong et al. maximized the expected utility of the energy consumption in a system with single energy source and multiple energy consumers. The stochastic energy demand constraints were turned into linear constraints using Gaussian approximation. And the size of the optimization problem was reduced using rolling horizon optimization. Lastly, the authors applied distributed Newton's method to decompose the optimization to decentralize the control system and the computational burdens. However, in order to get the optimal solution to the problem, approximately 50 rounds of variables update were required. As each round necessitated

several rounds of communication, high network reliability and computing power are essential to solve the problem.

C. Robust Optimization (RO)

RO, like SO, acknowledges the presence of uncertainty in the system. However, RO only assumes the parameters to lie within some bounds [22], which is more general than assuming known distribution for the parameters in SO. Historical data also help RO in updating the bounds for the parameters. Generally, RO is more suitable for risk-averse applications as it assumes the worst case scenario within the parameters' bound.

In [23], Zhang, et al. optimized the scheduling of a grid-connected microgrid with high wind energy penetration and demand side management (DSM). RO was used to minimize the net cost incurred by the microgrid – costs of distributed generation and storage subtracted by the utility of the electricity consumption. The authors assumed the utility function of the load to be increasing and concave, making the objective function convex. Dual decomposition was used to solve the algorithm iteratively and distributively. First, the values of the Lagrange multipliers were initialized by a centralized controller and were sent to local controllers, which would use those values to calculate optimum power generation and/or consumption values for the given Lagrange multipliers. The power generation and consumption schedules were then sent back to the centralized controller which would update the Lagrange multipliers using the subgradient method. The process continued until there was convergence in the power schedules. In the simulation, worst case wind power energy was considered. However, if forecasting was included, expectation values rather than worst case values could have been employed and resulted in a more realistic power dispatch. The authors did not mention the time required to solve the optimal scheduling either. As such, it is not known if the algorithm can be used in real time.

Wang, et al. [24] also used RO to minimize the operating cost of a grid-connected microgrid with renewable sources and combine heat and power (CHP) generators. Generation from RES was combined with the load to constitute the net demand. Then, reference distributions of the net and heat demand were formed based on historical data. The real net and heat demand distributions were allowed to fluctuate within a certain value around the reference distributions. Subsequently, chance constraint approximation and a robust approach were used to transform the demand constraints to linear constraints. Lastly, the problem was transformed into mixed integer linear programming (MILP) problem and was solved iteratively. The robust optimization was shown to perform 1-10% better than deterministic approach.

IV. HEURISTIC OPTIMIZATION

Many heuristic optimization methods have been developed and used to solve the optimization of power dispatch. Heuristic algorithms do not need the mathematical formulations of the variables as they generate solutions to the optimization problem directly and then make the solutions better in subsequent

iterations through different mechanisms. Heuristic optimization is categorized into population-based (e.g. Genetic Algorithm, Particle Swarm Optimization) and non-population-based (e.g. Tabu Search, Simulated Annealing).

A. Genetic Algorithm (GA)

GA is an optimization algorithm inspired by the evolution of organisms [25]. The algorithm is started by randomly generating a population of chromosomes, say 100 chromosomes. The fitness of each chromosome, which is a complete solution to the problem, is evaluated based on the objective function of the optimization problem. The chromosomes with the best fitness values are stored in a mating pool and are used to create a new generation of population using crossover operation. The new generation of chromosomes then follows the same iterative processes to produce the next generations of chromosomes with higher fitness value. The iterations continue until the stopping criteria are reached. The chromosomes may also be mutated to generate randomness and prevent the solutions from being trapped in local minima. The variables in implementing the algorithm include the population size, the crossover and mutation rate as well as the process to generate the mating pool. Very large mutation or crossover rate may cause GA to turn into a random search while very low rate may cause the solutions to be trapped in local minima. GA has been used widely in energy scheduling optimization. Flexibility and the scalability of GA are some of its advantages [26].

In [27], Yuce, et al. optimized real time scheduling of domestic appliances' usage in a four-bedroom house for a week at fifteen minutes time scale. The authors seek to minimize the energy taken from the grid, therefore maximizing the use of RES, whose availability was forecasted using ANN. Firstly, data for appliance operation patterns were generated using thermal energy modeling taking into account building materials and occupancy schedules among other information. Secondly, ANN was trained using the data to produce the energy consumption and the renewable energy generation scenario. Finally, GA was utilized to solve the optimization problem. Experiments were conducted to configure each of the parameters in ANN and GA. Grid energy reduction targets of 10%, 25% and 40% were simulated separately and the algorithm managed to schedule and align the household energy consumption with the availability of the RES to fulfill each of the targets.

Liang, et al. [28] also used GA to minimize the fuel cost of thermal generators subject to power balance and spinning reserve requirements. The system consisted of thermal, hydro, wind, solar and battery systems. Errors in the forecast were taken into account using fuzzy sets. Membership functions were created for each of the constraints, namely total fuel cost, load demand, spinning reserve requirement, available water, wind speed and solar radiation. These membership functions were then used to calculate the fitness value of the solutions. It was shown that by accounting for the uncertainties using fuzzy logic, the solution was 1% better than that of conventional GA, a reduction of \$72,000 per day in this case.

Mehta, et al. [29] designed and compared two charging strategies of electric vehicles (EV) based on technical and economic objective functions. The former's aim was to minimize the peak load while the latter's to minimize the total daily cost, considering charging cost and battery degradation cost. In optimizing the charging strategy, heuristic initialization was utilized to narrow down the search space. GA was then employed to formulate the charging strategy from the feasible search space. Maximum EV penetration for the two charging strategies were tested with voltage limits, transmission line limits, and rating of transformer as the constraints. As expected, the charging strategy that minimized peak load have higher maximum penetration of EV when simulated in a 37-bus distribution system. It was also found that the peak-load-minimizing strategy can delay the need for system improvement.

B. Particle Swarm Optimization (PSO)

Inspired by the collective behaviour of birds flocking or fish schooling, Kennedy and Eberhart first proposed PSO in 1995 [30]. As a population-based heuristic optimization, PSO is similar to GA, if not simpler. A population of particles is first generated and the location of each particle in the search space is a candidate solution. The fitness value of each of the particles is calculated and then the particles move with a certain velocity to a new location. The new locations for each of the particle are the new generation of solutions. The velocity of the particles depends on the best solution for a specific particle across generations and the best solution for all particles in that generation [31], [32].

Zhao, et al. [33] developed multiagent PSO (MAPSO) to optimize reactive power dispatch with the goal of minimizing active power losses in the system. The agent in the algorithm lives in a lattice-like environment and represents a particle in the PSO. The particles in MAPSO mimic the behaviour of agents in MAS, such as cooperating with or competing against each other, communicating with their neighbours, and learning through gathering knowledge. Because of the ability to interact with the neighbouring agents, information in a local agent lattice can diffuse slowly to the whole lattice. MAPSO was shown to converge much faster than the standard PSO and a few other algorithms. When simulated for 50 times in 30-bus and 118-bus transmission system, MAPSO achieved better consistency compared to standard GA and PSO.

Ziadi et al. [34] optimized the day-ahead scheduling of reactive power from PV, active and reactive power from battery energy storage system (BESS), active and reactive power demand from controllable loads (CL), tap changing transformers settings in a distribution system with high PV penetration. The optimized scheduling of PV, BESS, transformers and CL were compared with two other cases. One in which only tap positions of the transformers were optimized in the absence of PV, BESS and CL. The other case had PV, BESS and transformer settings optimized, but CL was not considered. Changing the tap positions of the transformers alone was not enough to keep the voltage level within the acceptable limit. The addition

of optimized PV and BESS not only managed to keep the voltage levels within the acceptable range, but also controlled the reverse power flow because of the high generation from PV. The introduction of CL then reduced the power losses in the system and decreased the size of BESS required. Similarly, [35] has also utilized PV and BESS to maintain the voltage levels of an off-grid hybrid system using PSO algorithm.

PSO has also been applied in optimization with discrete variables [36] by using binary PSO introduced in [37]. Authors of [36] controlled the state of water heater – either on or off to minimize electricity costs while maximizing users' comfort.

C. Tabu Search (TS)

TS uses a neighbourhood search procedure to move from one potential solution to an improved solution in the neighbourhood [38]. TS avoids being trapped in a local minima by using a memory structure called tabu list. Tabu list controls the search such that the algorithm will not explore the solutions that have been searched in the past n iterations. The quality of solution produced by TS has strong dependence on the initial solution.

Takeuchi, et al. [39] applied heuristics optimization to minimize CO₂ emissions and energy costs in Aichi, Japan and showed that the optimization worked in a real system. The power dispatch was optimized for a week such that the energy storage schedule could be configured for the different pattern between the weekdays and weekends. The energy control system first optimized the power scheduling a day in advance by using past performance data and weather forecast. Next, real time control was also applied, using phosphoric acid fuel cells which have fast response time, to compensate for error in forecasting. The authors compared TS with GA and found that the two algorithm's performance were very similar. TS performed slightly better on more days but were often trapped in the local minimum.

D. Simulated Annealing (SA)

The development of SA was inspired by the process of heating of a material and then cooling it down slowly to decrease its defects [40]. When compared to GA or PSO that generates large population in each iteration, simulated annealing requires much less computational time and resources.

In [41], Sousa, et al. considered the energy management of a smart grid with distributed generation, demand response and EV using Virtual Power Plant (VPP). The power generation technologies considered included PV, fuel cells, wind farms, micro hydro, and biomass. Seven types of EVs, different charging rates, and driving patterns were taken into account. The authors compared MILP with simulated annealing in minimizing the operation cost of the VPP. A 33-bus radial distribution system was used as a test system. The optimal solution obtained was then validated if it violated voltage constraints or current constraints at the nodes and branches respectively. While the computation time for SA was faster compared to MILP (0.8 seconds versus 1.4 seconds), MILP solution was slightly better (1.2%).

E. Other Heuristic Algorithms

There are many other optimization algorithms that are inspired by evolutionary characteristics and/or animal behaviours, such as differential evolution (DE) [42], evolutionary strategies (ES) [43], and ant colony optimization (ACO) [44]. However, they are not covered here because their applications in generation scheduling are not as widespread.

V. HYBRID ALGORITHM

Many researchers have also combined certain aspects of different algorithms, either from the same category or from both exact and heuristic approaches, to form hybrid algorithms that are customized to the purpose and the characteristics of the optimization problem.

Ji et al. [45] developed a hybrid inexact stochastic-fuzzy chance-constrained programming to account for vastly different types of uncertainties by synthesizing interval-parameter programming (IPP), two-stage stochastic programming (TSP) and fuzzy credibility constrained programming (FCCP). IPP and TSP together can handle uncertainty in power generation and demand in discrete intervals with lower and upper bound. Meanwhile, FCCP was employed to measure the confidence level of the availability of renewable energy sources, expressed in fuzzy sets. The authors used the algorithm to minimize the energy cost of a microgrid over 24 hours with four types of power generation technologies, namely coal, natural gas, PV and wind with the possibility of importing electricity from the grid. Simulations were carried out with varying confidence level, demand level, and emission reduction targets. The paper highlighted the trade-offs between the system economy, emission reduction and reliability in energy scheduling. The authors went on to advise which strategy should be adopted based on the objectives and risk tolerance of the system planner.

In [46], Kuznetsova, et al. optimized the individual objectives of microgrid stakeholders, each represented by an agent in a multiagent platform. The system consisted of a train station that plays the role of consumer and producer, a surrounding district and a small wind power plant which can sell and buy power from each other. The district's aim was to minimize its power purchase costs, wind plant's to maximize revenue from supplying power and station's to minimize net power costs. The unpredictable RES availability and load were quantified as Prediction Intervals (PIs) using Non-dominated Sorting Genetic Algorithm (NSGA-II)-trained ANN. The trained ANN generated PIs that were optimized in terms of coverage probability and the prediction interval width. Possibility of generation failure and the subsequent penalty were also taken into account in the cost function of the wind plant. The algorithm was then tested in offline mode. When compared with a deterministic-optimized system, the robust-optimized system suffers from up to 60 times less shortage, albeit earning 4 times less revenue.

Within heuristic algorithms, PSO has been hybridized with ES to develop optimum charging strategy [47]. The authors used fuzzy logic based stochastic model to generate driving patterns of the EVs and PSO was used to find the parameters

of the membership functions. To design the charging strategy, ES was first utilized to find an optimal V2G strategy. Then, the optimal charging sequence was found using PSO. The vast search space was managed in a shorter time by using ES.

VI. DISCUSSION

A. Algorithm Perspective

Both exact and heuristic optimizations have been applied successfully in a wide range of cases, from household level to transmission level. It is observed that uncertainties can be more easily integrated in exact optimization, which is why forecasting is more frequently utilized in exact optimization than in heuristic approach. In heuristic algorithms, the uncertainties are more often dealt using fuzzy logic [11], [28].

In general, if we can solve a problem using exact optimization approach, i.e. the mathematical expressions for the objective functions and the constraints are known and can be solved in polynomial time without approximations that will change the nature of the problem, then we should use the exact optimization. However, it is often the case that the exact optimization problem is NP-hard, or that the constraints cannot be expressed analytically yet. This is most notable in the case of EVs where EVs availability, driving patterns, user preference and other constraints are intractable. In these cases, heuristic optimizations have been successfully employed. Even though heuristic optimization can never guarantee that the solutions it gives are the optimum solution, they are often good enough and can be obtained in a reasonable time scale.

That being said, when the formulation is tractable, heuristic optimizations typically require higher computational time compared to its exact counterpart. Therefore, heuristic optimizations are more applicable when used to solve planning problem such as day ahead generation scheduling or sizing and siting of DERs, rather than to solve a real time power dispatch.

There are increasingly many hybrid algorithms, whether it be algorithms that incorporate different aspects of heuristic approaches, such as PSO with mutation and crossover operator [48], [49], or algorithms that incorporate both elements from exact and heuristic optimization to enjoy the advantages of the different techniques [46].

B. System Perspective

Almost all the optimization approaches reviewed are either solely dealing with the optimization of active power (P) or exclusively handling the reactive power (Q) dispatch. On the one hand, most of the active power optimizations do not consider the system configuration, line impedances, voltage stability, and other power system constraints [16], [18], [21], [23], [24], [28]. Their focus lies on the minimization of the cost. On the other hand, the reactive power optimizations do not take into account the economic aspects, assume P generation is constant, and only deal with the voltage stability and power losses [20], [33], [47]. Only the authors of [34] have considered both P and Q optimization simultaneously. Nonetheless, the

costs of producing Q have not been taken into account. This is understandable as the constraints from the power system increase the complexity of the P optimization while the cost of Q dispatch is much lower compared to the cost of P and has not been fully understood. Most of the works also neglected the degradation of battery, even though battery may be an integral part of the problems. Therefore, for the system to be more realistic, there needs to be more work whereby both P and Q are optimized simultaneously while considering the system constraints, reactive power costs, as well as battery degradation constraints among other practical constraints [31].

Many authors have also solved the optimization problem distributively [16], [21], [23], [46]. And this trend will continue as DERs are becoming more common and control is getting more decentralized. In this aspect, MAS is a very powerful tool that allows the different components in the grid to have their own objectives. More works will be integrated into the MAS platform in the future.

Last but not least, while the optimization approaches have shown tremendous benefits for the systems in which they are employed, these benefits are still confined to the simulation playground. There is an increasing need to apply these optimizations to real systems to understand their usefulness better and to understand the practical constraints of a real system more comprehensively. After all, the optimization algorithms were developed and will continue to be developed by researchers to solve real world problems.

VII. CONCLUSION

With the increasing adoption of non-dispatchable renewable energy sources such as solar and wind, as well as the growing presence of electric vehicles in power system, optimizations of power dispatch is more important than ever. This paper has given an overview of the different types of optimization tools and algorithms utilized in power dispatch. Exact and heuristic optimization have been compared and contrasted and their applications in the literature have been presented and summarized. In the future, it is predicted that more hybrid algorithms will be employed to solve optimization problems in test systems that will become increasingly more realistic.

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REFERENCES

- [1] Y. M. Atwa, E. F. El-Saadany, M. M. A. Salama, and R. Seethapathy, "Optimal Renewable Resources Mix for Distribution System Energy Loss Minimization," *Power Systems, IEEE Transactions on*, vol. 25, no. 1, pp. 360–370, 2010.
- [2] C. Breyer and A. Gerlach, "Global overview on grid-parity," *Progress in Photovoltaics: Research and Applications*, vol. 21, pp. 121–136, jan 2013.
- [3] International Energy Agency, "World Energy Outlook 2013," tech. rep., 2013.
- [4] B. Saravanan, S. Das, S. Sikri, and D. P. Kothari, "A solution to the unit commitment problem-a review," *Frontiers in Energy*, vol. 7, no. 2, pp. 223–236, 2013.

Table I
SUMMARY OF LITERATURE SURVEYED

Ref.	Optimization tools	Objective function	T/D/H ^a	Control variables in the system
[16]	Distributed DO (ADMM, MPC)	Minimize P taken from the grid	D	P of PV, BESS, EV and CL
[17]	DO (LP)	Minimize P taken from the grid	H	P of BESS, BESS capacity, peak load reduction target
[18]	offline and online DO (convex)	Minimize P taken from the grid	D	P of BESS, P from the grid
[20]	SO (SOCP)	Minimize P loss	D	Q of PV, subject to stochastic P of PV
[21]	Distributed SO	Maximize expected utility of energy consumption	D	P of background load and 2 types of dynamic load
[23]	Distributed RO (convex)	Minimize total cost of a microgrid	D	P of BESS, CL and P from the grid
[24]	RO	Minimize total cost of a microgrid	D	Energy from CHP generators and P from the grid
[26]	GA	Minimize peak load demand	D	28 types of equipment's schedule
[27]	ANN and GA	Minimize P from the grid	H	9 types of household appliances' schedule
[28]	Fuzzy logic and GA	Minimize fuel cost of thermal generators	T	P of thermal generator, volume of water released for hydro units, wind speed, solar irradiance
[29]	GA	Minimize total cost of charging; minimize peak load	D	charging sequence of EVs (P), EV penetration
[31]	PSO	Minimize total cost of a distribution system	D	Q of PV, Q of EV
[33]	Multiagent PSO	Minimize P loss	T	generator bus voltages, tap positions of transformer, reactive power source installation
[34]	PSO	Minimize P loss	D	Q of PV, P and Q of BESS, CL, tap positions of transformer
[36]	Binary PSO	Minimize peak load demand and maximize water temperature	H	water heater position (on/off)
[39]	TS	Minimize CO ₂ emission or energy costs	D	P of thermal generators, BESS, and 3 types of fuel cells
[41]	SA	Minimize total cost of a VPP	D	P of CL, fuel cell, and 7 types of EV, P from the grid
[42]	DE	Minimize total cost of thermal generators	T	P of EV, P of thermal generators
[43]	ES	Minimize P loss	T	generator bus voltages, tap positions of transformer, Q of capacitor banks
[44]	ACO	Minimize P loss	T	generator bus voltages, tap positions of transformer, Q of capacitor banks
[45]	Stochastic, Fuzzy, Chance-constrained Programming	Minimize total cost of a microgrid	T	P of coal and gas plants, P from grid, confidence level, demand level, emission reduction targets,
[46]	MAS, GA, ANN, RO	Minimize total cost of a district; maximize revenue from supplying P ; minimize net costs	D	P from the grid, train station, and from wind plant P from/to the grid, and P to the district P from/to the grid, P to the district, P of BESS, CL
[47]	ES PSO	Minimize costs of EV for V2G services	D	charging sequence of EVs (P)

^aThe letters T,D, and H indicate whether the optimization happens at Transmission system, Distribution system, or Household level respectively. Distribution system also include the category of microgrid because of the similar size and similar variables considered.

- [5] R. Baños, F. Manzano-Agugliaro, F. G. Montoya, C. Gil, A. Alcayde, and J. Gómez, "Optimization methods applied to renewable and sustainable energy: A review," *Renewable and Sustainable Energy Reviews*, vol. 15, no. 4, pp. 1753–1766, 2011.
- [6] F. Rosenblatt, "The perceptron: a probabilistic model for information storage and organization in the brain," *Psychological review*, vol. 65, no. 6, pp. 386–408, 1958.
- [7] F. Wang, Z. Mi, S. Su, and H. Zhao, "Short-term solar irradiance forecasting model based on artificial neural network using statistical feature parameters," *Energies*, vol. 5, no. 5, pp. 1355–1370, 2012.
- [8] B. Yuce, H. Li, Y. Rezgui, I. Petri, B. Jayan, and C. Yang, "Utilizing artificial neural network to predict energy consumption and thermal comfort level: An indoor swimming pool case study," *Energy and Buildings*, vol. 80, pp. 45–56, 2014.
- [9] J. F. Tang, D. W. Wang, R. Y. K. Fung, and K.-L. Yung, "Understanding of fuzzy optimization: theories and methods," *Journal of Systems Science and Complexity*, vol. 17, no. 1, pp. 117–136, 2004.
- [10] Z. Kovacic and S. Bogdan, *Fuzzy Controller Design: Theory and Applications*, vol. 19 of *Automation and Control Engineering*. CRC Press, dec 2006.
- [11] B. M. Radhakrishnan and D. Srinivasan, "A multi-agent based distributed energy management scheme for smart grid applications," *Energy*, vol. 103, pp. 192–204, may 2016.
- [12] F. Bellifemine, G. Caire, and D. Greenwood, *Developing Multi-Agent with JADE Systems*. 2007.
- [13] A. Kantamneni, L. E. Brown, G. Parker, and W. W. Weaver, "Survey of multi-agent systems for microgrid control," *Engineering Applications of Artificial Intelligence*, vol. 45, pp. 192–203, 2015.
- [14] A. R. Khare and B. Y. Kumar, "Multiagent structures in hybrid renewable power system: A review," *Journal of Renewable and Sustainable Energy*, vol. 7, no. 6, p. 063101, 2015.
- [15] M.-H. Lin, J.-F. Tsai, and C.-S. Yu, "A Review of Deterministic Optimization Methods in Engineering and Management," *Mathematical Problems in Engineering*, vol. 2012, pp. 1–15, 2012.
- [16] T. Wang, D. O'Neill, and H. Kamath, "Dynamic Control and Optimization of Distributed Energy Resources in a Microgrid," *IEEE Transactions on Smart Grid*, vol. 6, pp. 2884–2894, nov 2015.
- [17] A. Nottrott, J. Kleissl, and B. Washom, "Energy dispatch schedule optimization and cost benefit analysis for grid-connected, photovoltaic-battery storage systems," *Renewable Energy*, vol. 55, pp. 230–240, 2013.
- [18] K. Rahbar, J. Xu, and R. Zhang, "Real-Time Energy Storage Management for Renewable Integration in Microgrid: An Off-Line Optimization Approach," *IEEE Transactions on Smart Grid*, vol. 6, no. 1, pp. 124–134, 2015.
- [19] D. Fouskakis and D. Draper, "Stochastic Optimization: A Review," *International Statistical Review*, vol. 70, no. 3, pp. 315–349, 2002.
- [20] V. Kekatos, G. Wang, A. J. Conejo, and G. B. Giannakis, "Stochastic Reactive Power Management in Microgrids With Renewables," *IEEE Transactions on Power Systems*, vol. PP, no. 99, pp. 1–10, 2014.
- [21] C. Gong, X. Wang, W. Xu, and A. Tajer, "Distributed Real-Time Energy Scheduling in Smart Grid: Stochastic Model and Fast Optimization," *IEEE Transactions on Smart Grid*, vol. 4, no. 3, pp. 1476–1489, 2013.
- [22] H. G. Beyer and B. Sendhoff, "Robust optimization - A comprehensive survey," *Computer Methods in Applied Mechanics and Engineering*, vol. 196, no. 33–34, pp. 3190–3218, 2007.
- [23] Y. Zhang, N. Gatsis, and G. B. Giannakis, "Robust energy management for microgrids with high-penetration renewables," *IEEE Transactions on Sustainable Energy*, vol. 4, no. 4, pp. 944–953, 2013.
- [24] R. Wang, P. Wang, and G. Xiao, "A robust optimization approach for energy generation scheduling in microgrids," *Energy Conversion and Management*, vol. 106, pp. 597–607, 2015.
- [25] S. Sivanandam and S. Deepa, *Introduction to Genetic Algorithms*. Springer, 2008.
- [26] T. Logenthiran, D. Srinivasan, and T. Z. Shun, "Demand side management in smart grid using heuristic optimization," *IEEE Transactions on Smart Grid*, vol. 3, no. 3, pp. 1244–1252, 2012.
- [27] B. Yuce, Y. Rezgui, and M. Mourshed, "ANN-GA smart appliance scheduling for optimised energy management in the domestic sector," *Energy and Buildings*, vol. 111, pp. 311–325, 2016.
- [28] R.-H. Liang and J.-H. Liao, "A Fuzzy-Optimization Approach for Generation Scheduling With Wind and Solar Energy Systems," *IEEE Transactions on Power Systems*, vol. 22, no. 4, pp. 1665–1674, 2007.
- [29] R. Mehta, D. Srinivasan, A. M. Khambadkone, J. Yang, and A. Trivedi, "Smart Charging Strategies for Optimal Integration of Plug-in Electric Vehicles within Existing Distribution System Infrastructure," *IEEE Transactions on Smart Grid*, vol. 3053, no. c, pp. 1–1, 2016.
- [30] J. Kennedy and R. Eberhart, "Particle swarm optimization," *Neural Networks, 1995. Proceedings., IEEE International Conference on*, vol. 4, pp. 1942–1948 vol.4, 1995.
- [31] O. Gandhi, W. Zhang, C. D. Rodríguez-Gallegos, D. Srinivasan, and T. Reindl, "Continuous Optimization of Reactive Power from PV and EV in Distribution System," in *2016 IEEE Innovative Smart Grid Technologies - Asia (ISGT Asia)*, (Melbourne, 2016. (To be published)).
- [32] C. D. Rodríguez-Gallegos, K. Rahbar, M. Bieri, O. Gandhi, T. Reindl, and S. K. Panda, "Optimal PV and Storage Sizing for PV-Battery-Diesel Hybrid Systems," in *Industrial Electronics Society, 2016. IECON 2016. 42nd Annual Conference of IEEE*, (Florence, 2016. (To be published)).
- [33] B. Zhao, C. Guo, and Y. Cao, "A Multiagent-Based Particle Swarm Optimization Approach for Optimal Reactive Power Dispatch," *IEEE Transactions on Power Systems*, vol. 20, no. 2, pp. 1070–1078, 2005.
- [34] Z. Ziadi, S. Taira, M. Oshiro, and T. Funabashi, "Optimal power scheduling for smart grids considering controllable loads and high penetration of photovoltaic generation," *IEEE Transactions on Smart Grid*, vol. 5, no. 5, pp. 2350–2359, 2014.
- [35] C. D. Rodríguez-Gallegos, M. S. Alvarez-Alvarado, O. Gandhi, D. Yang, W. Zhang, T. Reindl, and S. K. Panda, "Placement and Sizing Optimization for PV-Battery-Diesel Hybrid Systems," in *IEEE 4th International Conference on Sustainable Energy Technologies (ICSET)*, (Hanoi, 2016. (To be published)).
- [36] A. Sepulveda, L. Paull, W. G. Morsi, H. Li, C. P. Diduch, and L. Chang, "A novel demand side management program using water heaters and particle swarm optimization," *2010 IEEE Electrical Power & Energy Conference*, pp. 1–5, 2010.
- [37] J. Kennedy and R. Eberhart, "A discrete binary version of the particle swarm algorithm," *1997 IEEE International Conference on Systems, Man, and Cybernetics. Computational Cybernetics and Simulation*, vol. 5, pp. 4–8, 1997.
- [38] F. Glover and M. Laguna, *Tabu Search*. Boston, MA: Springer US, 1997.
- [39] A. Takeuchi, T. Hayashi, Y. Nozaki, and T. Shimakage, "Optimal scheduling using metaheuristics for energy networks," *IEEE Transactions on Smart Grid*, vol. 3, no. 2, pp. 968–974, 2012.
- [40] P. J. M. V. Laarhoven and E. H. L. Aarts, *Simulated annealing: theory and applications*. 1987.
- [41] T. Sousa, H. Morais, Z. Vale, P. Faria, and J. Soares, "Considering Vehicle-to-Grid : A Simulated Annealing Approach," *IEEE Transactions on Smart Grid*, vol. 3, no. 1, pp. 535–542, 2012.
- [42] D. S. Kumar, D. Srinivasan, and T. Reindl, "Optimal power scheduling of distributed resources in Smart Grid," in *2013 IEEE Innovative Smart Grid Technologies-Asia (ISGT Asia)*, vol. 09, pp. 1–6, IEEE, nov 2013.
- [43] D. B. Das and C. Patvardhan, "A new hybrid evolutionary strategy for reactive power dispatch," *Electric Power Systems Research*, vol. 65, no. 2, pp. 83–90, 2003.
- [44] A. Abbasy and S. Hosseini, "Ant Colony Optimization-Based Approach to Optimal Reactive Power Dispatch: A Comparison of Various Ant Systems," *2007 IEEE Power Engineering Society Conference and Exposition in Africa - PowerAfrica*, no. July, pp. 16–20, 2007.
- [45] L. Ji, D. Niu, M. Xu, and G. Huang, "An optimization model for regional micro-grid system management based on hybrid inexact stochastic-fuzzy chance-constrained programming," *International Journal of Electrical Power & Energy Systems*, vol. 64, pp. 1025–1039, 2015.
- [46] E. Kuznetsova, Y. F. Li, C. Ruiz, and E. Zio, "An integrated framework of agent-based modelling and robust optimization for microgrid energy management," *Applied Energy*, vol. 129, pp. 70–88, 2014.
- [47] J. Tan and L. Wang, "Integration of plug-in hybrid electric vehicles into residential distribution grid based on two-layer intelligent optimization," *IEEE Transactions on Smart Grid*, vol. 5, no. 4, pp. 1774–1784, 2014.
- [48] V. Miranda and N. Fonseca, "EPSO-evolutionary particle swarm optimization, a new algorithm with applications in power systems," *IEEE/PES Transmission and Distribution Conference and Exhibition*, vol. 2, pp. 745–750, 1995.
- [49] D. Li, Liquan Gao, Junzheng Zhang, and Yang Li, "Power System Reactive Power Optimization Based on Adaptive Particle Swarm Optimization Algorithm," in *2006 6th World Congress on Intelligent Control and Automation*, vol. 14, pp. 7572–7576, IEEE, 2006.